RV COLLEGE OF ENGINEERING
Bengaluru-560 059

REPORT ON

PROJECT BASED LEARNING

ACY 2024-25

Title of the Project

4-Bit Booth Encoded Wallace Tree Multiplier

Students Group

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1.	
2.	

1. Introduction

Multiplication is one of the most fundamental arithmetic operations in digital systems and plays a critical role in processors, digital signal processors (DSPs), graphics hardware, and machine learning accelerators. As system performance requirements increase, there is a strong need for high-speed and hardware-efficient multiplier architectures.

Conventional array multipliers suffer from large propagation delays due to serial addition of partial products. To overcome these limitations, advanced techniques such as **Booth encoding** and **Wallace tree reduction** are widely used. Booth encoding reduces the number of partial products, while Wallace tree structures compress partial products in parallel, significantly improving speed.

This project focuses on the **design and implementation of a 4-bit Booth-encoded Wallace-Tree Multiplier**, combining Modified Radix-4 Booth encoding with Wallace tree compression. The design is described using hardware description language and verified through simulation and FPGA implementation.

2. Literature Review

Several researchers have explored optimized multiplier architectures to improve speed, area efficiency, and power consumption.

Sithara *et al.* (2025) proposed a Modified Booth–Wallace Tree multiplier that significantly reduces partial products and minimizes carry propagation delay. Their FPGA implementation demonstrated improved timing performance, making the design suitable for high-speed image processing and DSP applications.

Baqi *et al.* (2025) presented a comparative study between Booth-Wallace and Vedic multipliers. The results showed that Booth-Wallace architectures achieve higher speed due to parallel reduction, while Vedic multipliers are more area-efficient. This highlights the trade-off between speed and hardware utilization.

Wang (2024) analyzed an 8-bit Wallace Tree multiplier using carry-save adders and a ripple carry adder in the final stage. The study demonstrated that limiting carry propagation to the final stage significantly improves computational speed while maintaining reasonable area complexity.

These studies confirm that Booth encoding combined with Wallace tree reduction is an effective approach for designing high-speed multipliers.

3. Problem Statement

Traditional multipliers based on array architectures suffer from increased delay due to sequential carry propagation. As operand size increases, the number of partial products and addition stages grows rapidly, limiting performance.

The problem addressed in this project is the design of a fast and efficient 4-bit multiplier that:

- Reduces the number of partial products
- Minimizes carry propagation delay

- Achieves correct signed multiplication
- Is suitable for FPGA implementation

4. Objectives

4.1 Primary Objectives

- To implement Modified Radix-4 Booth encoding for partial product generation
- To design a Wallace tree structure using carry-save adders for fast partial product reduction
- To generate an accurate 8-bit product from two 4-bit signed inputs
- To verify functionality through simulation and FPGA synthesis

4.2 Secondary Objectives

- To analyze timing and power performance using FPGA tools
- To demonstrate scalability towards higher bit-width multipliers
- To study the practical implementation of high-speed arithmetic circuits

4. Methodology

The multiplier design follows a structured methodology combining algorithmic optimization and hardware realization.

5.1 Booth Partial Product Generation

- A zero is appended to the least significant bit of the multiplier
- Overlapping 3-bit groups are formed for Radix-4 Booth recoding
- Each group generates one of the following operations: 0, +A, −A, +2A, −2A
- Partial products are shifted and sign-extended based on group position

5.2 Wallace Tree Reduction

- Partial products are aligned according to bit weight
- Full adders (3:2 compressors) and half adders (2:2 compressors) are used
- Columns are reduced in parallel until only two rows remain

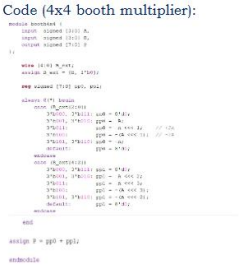
5.3 Final Addition

- The remaining two rows are added using a fast adder
- The final 8-bit product is generated

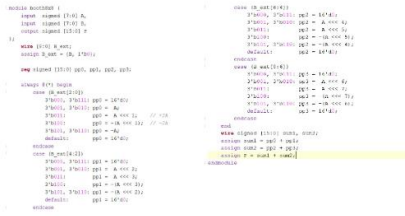
5.4 Verification

- Functional verification is performed using simulation tools
- The design is synthesized and tested on an FPGA platform

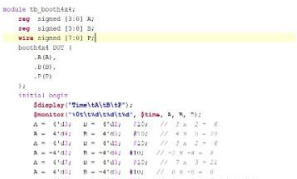
Code (4x4 booth multiplier):



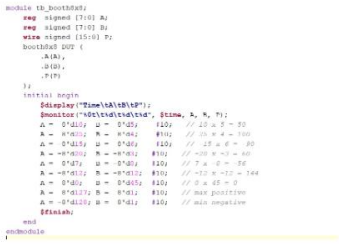
Code (8x8 booth multiplier):

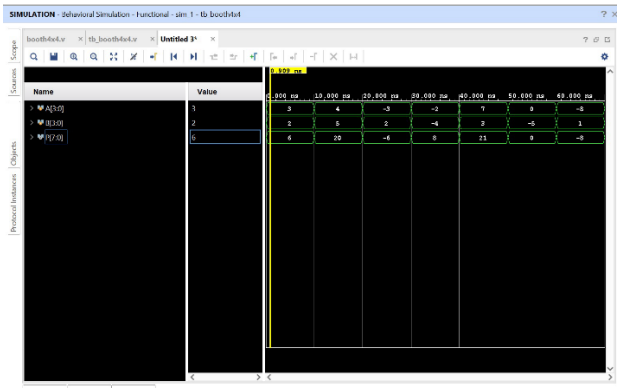


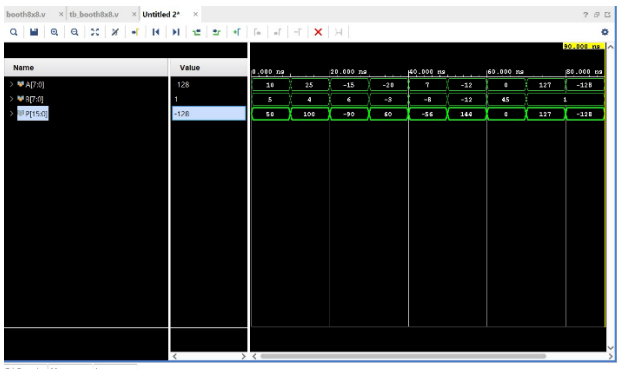
Testbench for 4x4:



Testbench for 8x8:







6. Hardware and Software Requirements

<u>Hardware</u>

- Laptop
- FPGA board (PYNQ-Z1)

Software

- Xilinx Vivado / ModelSim for simulation and synthesis

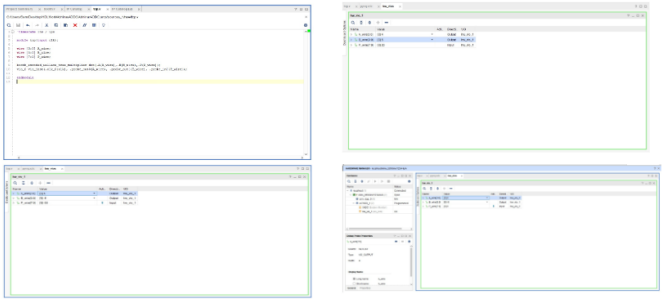
7. Implementation

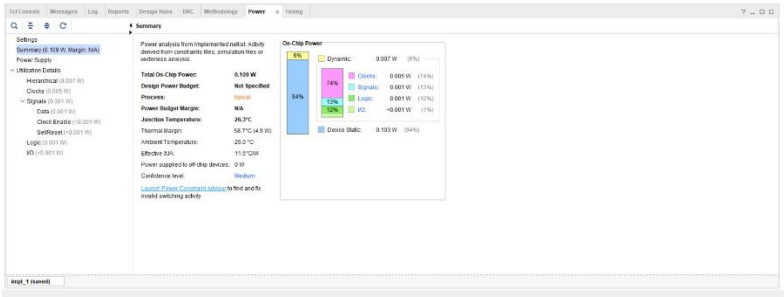
The multiplier was implemented using hardware description language with separate modules for Modified Booth encoding, partial product generation, Wallace tree reduction, and final addition.

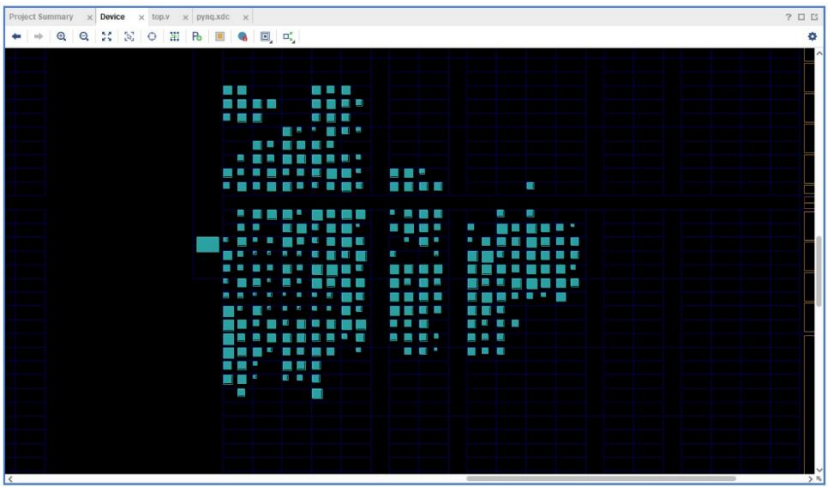
Both **4×4 and 8×8 Booth multipliers were simulated** using appropriate testbenches to verify correct signed multiplication for positive and negative operands. The simulation results confirmed functional correctness and validated the design logic.

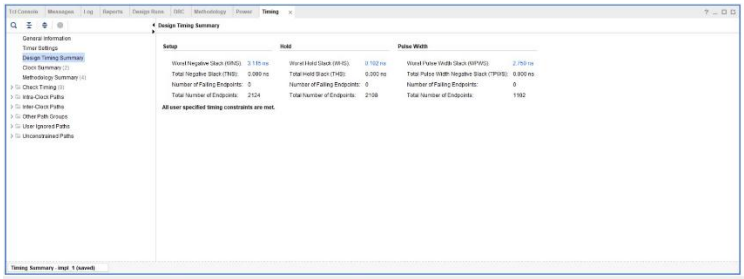
However, **only the 4×4 Booth-encoded Wallace-Tree multiplier was synthesized and implemented on FPGA hardware**. The 8×8 multiplier was limited to simulation due to increased resource requirements and was used to demonstrate scalability of the proposed architecture.

Virtual input/output code and output:









8. Corrected Results and Discussion

Simulation waveforms for both 4×4 and 8×8 multipliers showed correct multiplication results across all tested input combinations. This verified the correctness of the Booth encoding and Wallace tree reduction methodology.

FPGA implementation was successfully carried out **only for the 4×4 multiplier**, and hardware verification confirmed reliable operation on the FPGA platform. Timing analysis showed that the design met all specified constraints, and power analysis indicated low on-chip power consumption.

The 8×8 multiplier, although not implemented on FPGA, demonstrated through simulation that the architecture is scalable to higher bit widths.

9. Conclusion

This project successfully demonstrates the design and simulation of a Booth-encoded Wallace-Tree multiplier and the hardware implementation of a 4-bit multiplier on FPGA. Modified Radix-4 Booth encoding was used to reduce the number of partial products, while the Wallace tree structure enabled fast parallel reduction, significantly improving multiplication speed compared to conventional architectures.

Both 4×4 and 8×8 multipliers were functionally verified through simulation, confirming correct signed multiplication for all tested input combinations. The 4×4 multiplier was synthesized and implemented on FPGA, where it demonstrated reliable operation, met timing constraints, and exhibited low on-chip power consumption. The 8×8 multiplier was limited to simulation and served to validate the scalability of the proposed architecture.

Overall, the project highlights the effectiveness of combining Booth encoding with Wallace tree reduction for high-speed arithmetic applications. The successful FPGA implementation of the 4-bit design establishes a strong foundation for extending the architecture to higher bit-width multipliers and advanced digital signal processing systems in future work.

List of References:

[1] J. S. Sithara, A. Alfred Kirubaraj, and S. Senith, "Implementation of modified Booth–Wallace tree multiplier for image processing," in *Proc. Int. Conf. on Advanced Research in Electronics and Communication Systems (ICARECS)*, Atlantis Highlights in Engineering, pp. 258–273, 2025.

[2] N. A. A. S. Baqi, A. S. Mokhtar, M. T. Jusoh@Tajudin, and S. N. Mohd Tawil, "Performance comparison of Booth-Wallace and Vedic multiplier design," *Jurnal Kejuruteraan*, vol. 37, no. 6, pp. 2627–2635, 2025.

[3] P. T. Saiteja and J. Femila Roseline, "Design and implementation of 8-bit Wallace multiplier with reduced complexity using ripple carry adder in comparison with Sklansky adder," *AIP Conference Proceedings*, vol. 2853, no. 1, p. 020048, 2024.